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MILITARY SPECIFICATION

ALTIMETER SET, ELECTRONIC AN/APN-141(V)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide an accurate indication of absolute altitude of an aircraft over land or water.

1.2 Classification - The Electronic Altimeter Set shall consist of the items listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.10 of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

*3.2 Parts and Material - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4.)

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.2.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied as outlined in Specification MIL-E-4682.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

*3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed the sum of the individual weights specified for the units used.

FSC 5826

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3.3.2 Reliability

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance; continuously or intermittently for a period of at least 50 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a minimum total operating life of 3000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

*3.3.2.3 Reliability in Mean Time Between Failures (MTBF) - For the test program under 4.4.3 the mean (operating) time between failures shall be 400 hours.

3.3.2.4 Time Totalizing Meter - The equipment shall contain time totalizing meters in accordance with Specification MIL-M-7793. Meters conforming to MS17322 and having a maximum range of 10,000 hours shall be provided for both the Receiver-Transmitter RT-601B/APN-141(V) and the RF Switching Unit SA-791A/APN-141(V).

3.3.3 Cabling and Connections

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

*3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ±1.0 V AC and 5 ±0.5 V AC or DC

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3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 3 equipment, except as modified herein.

3.3.9.1 Vibration - When normally mounted, the equipment shall operate satisfactorily when subjected to continuous vibration within the frequency range of 5 to 500 cps and having a double amplitude of 0.06 inch or a maximum acceleration of ±10 g, whichever is the limiting value.

3.3.9.1.1 Extended Vibration Requirement - The Antennas, Receiver-Transmitter, RF Switching Unit and Coupler shall operate satisfactorily when subjected to continuous vibration in the vertical direction (as normally mounted) within the frequency ranges of 500 to 1,000 cps at an acceleration no less than ±4 g and 1,000 to 2,000 cps at an acceleration no less than ±2 g. The Antennas, Receiver-Transmitter, RF Switching Unit and Coupler shall operate satisfactorily when subjected to continuous vibration in a lateral direction (as normally mounted) within the frequency range of 500 to 2,500 cps at an acceleration of no less than ±2 g. For purposes of this specification "continuous operation" shall mean a vibration sweep rate no greater than 8 $\sqrt{f}$ cps per minute, where f is cycles per second of applied vibration.

3.3.9.2 Temperature - Altitude

(1) Receiver-Transmitter, RF Switching Unit, Antennas, and Coupler - Continuous operation over the temperature range of -54°C to +95°C, the temperature remaining constant or varying at a rate as high as 1°C per second.

(2) Indicator - Continuous operation over the temperature range of -54°C to +71°C, the temperature remaining constant or varying at a rate as high as 1°C per second.

(3) Receiver-Transmitter, RF Switching Unit, Antennas, Indicator and Coupler - Continuous operation at barometric pressures ranging from 30 inches Hg down to 0.81 inches Hg (approximately 80,000 feet altitude) the pressure remaining constant for long periods of time or varying at a rate as high as 0.5 inch Hg per second.

(4) Receiver-Transmitter, RF Switching Unit, Antennas and Coupler - Continuous operation over the temperature altitude range as defined by Figure 1, Sheet 3, of MIL-T-5422; the temperature and altitude may vary at a rate as high as 1°C per second and 0.5 inch Hg per second, respectively.

(5) Indicator - Continuous operation over the temperature altitude range as defined by Figure 1, Sheet 2, of MIL-T-5422; the temperature and altitude may vary at a rate as high as 1°C per second and 0.5 inch Hg per second, respectively.

3.3.9.3 Shock - Equipment shall not suffer damage or subsequently fail to provide the performance herein specified when subjected to 18 impact shocks of 15 g having a time duration of 11 ±1 milliseconds; the maximum "g" shall occur at approximately 5-1/2 milliseconds. The Receiver-Transmitter, RF Switching Unit and Coupler must survive shock loads of 32 g when in operation. Operation during the period of shock application is not required, but must be resumed with no reduction in accuracy in less than 1.0 second after cessation of the shock load.

*3.3.9.4 Humidity - Equipment shall not suffer damage or deterioration or subsequently fail to provide the performance herein specified when subjected to an environment of relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of water and frost.

3.3.9.5 Acoustical Noise - When the Antennas, Receiver-Transmitter, RF Switching Unit and Coupler are normally mounted, they shall operate satisfactorily when subjected to the following acoustical environment for a period of three hours:

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127.5 db in 75-150 cps band

126.0 db in 150-300 cps band

119.5 db in 300-600 cps band

119.5 db in 600-1200 cps band

123.0 db in 1200-2400 cps band

117.0 db in 2400-4800 cps band

113.0 db in 4800-9600 cps band

132.5 db overall

3.3.9.6 Equipment Nonoperation - Equipment shall not suffer damage or subsequently fail to provide the performance herein specified when subjected to the following storage environment.

3.3.9.6.1 Temperature - Temperature ranging from -62°C to +125°C for all components except the Indicator. Temperature ranging from -62°C to +95°C for Indicator only. Temperature shock of 125°C delta temperature for all components.

3.3.10 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 5 minutes at extreme service conditions.

*3.3.11 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources. The power required shall not exceed the specified amounts.

(1) AC power (single phase) 115 V, 400 cps, category "B" 90 volt-amperes.

(2) 5.0 volts, AC or DC, 1.0 amp (for Indicator lighting)

The electric power described above shall also be sufficient to operate the equipment as necessary to perform all servicing functions on the equipment.

3.3.11.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The Electronic Altimeter Set is a range tracking radar that utilizes pulse techniques and leading edge tracking of ground return signals to produce highly accurate altitude information from 0 to 5,000 feet of altitude. The altimeter system has two modes of operation. In the search mode, the system successively examines increments of range with each cycle of operation until the complete altitude range is searched for a ground return signal. In the track mode, the system locks on and tracks a ground return signal giving continuous altitude information. Altitude information is obtained by measuring the time from the instant that r-f energy leaves the transmitter until the first ground return signal is detected in the receiver. The system converts this time interval information to a dc voltage. This voltage is applied to a device which indicates altitude (in feet) proportional to the amplitude of the dc voltage.

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3.4.2 Altitude Range - The equipment shall operate reliably and indicate absolute altitude from 0 to 5,000 feet over water or land.

3.4.3 Accuracy - The over-all system accuracy shall be less than ±5 feet or ±5 percent of the correct terrain clearance, whichever is the greater.

3.4.4 Frequency - The equipment shall operate within the allocated frequency range of 4200 to 4400 megacycles with a center frequency of 4300 ±10 megacycles and with guard bands at both ends of the range to preclude interference with equipment operating outside this range.

3.4.5 Electrical Tracking Rate

*3.4.5.1 Indicated Altitude - The equipment including Indicator shall follow changes in height at the rate of not less than A/2 feet/second or 50 feet/second whichever is greater (where A equals the indicated altitude in feet).

*3.4.5.2 Altitude Signal - The altitude signal (equipment less the Indicator) shall follow changes in height at the rate of not less than 400 feet/second below 400 feet of indicated altitude and 3000 feet/second above 400 feet of indicated altitude. Tracking error shall be less than 2% of tracking rate below 400 feet and less than 1/2% of tracking rate above 400 feet.

*3.4.6 Ground Speed Limit - The system shall provide reliable operation over level terrain and constant aircraft height at all ground speeds up to 2000 knots.

3.4.7 Minimum Reflected Signal - Minimum reflected signal at 5,000 feet shall be no less than 9 db above that required for reliable operation assuming 12 db attenuation at reflection.

3.4.8 Aircraft Attitude - The system shall operate normally and reliably in an aircraft when climbing and diving at 50° angles and when banking at 30° angles. No detrimental effects shall be caused to the operation of the equipment by any flight maneuver, and operation of the equipment in any position shall not prevent normal operation when the aircraft returns to normal flight.

3.4.9 False Indication - The system shall not lock on noise or any interfering or false signal.

*3.4.10 Low Altitude Limit Function - A switching function shall be provided in the altimeter which will complete an electrical circuit when the absolute altitude is lower than the preselected height. The switching function in the ID-881 and ID-1015 units shall close and open at the limit index setting within +5 feet or +5 percent of the indicated altitude, whichever is greater (±10 feet or ±10 percent of the existing altitude). The switching function in the ID-1304 units shall close and open at the limit index setting ±10 feet or 5 percent of the indicated altitude, whichever is greater (±15 feet or ±10 percent of the existing altitude). The low altitude warning circuit shall not operate when (1) the signal strength becomes inadequate or (2) the system is turned off or (3) thealtitude is above 5,000 feet or (4) the indicated altitude is zero feet and the limit index is set to the minimum altitude to which it can be set without actuating the ON-OFF switch to the OFF position. The circuit shall be capable of carrying and switching a two-ampere lamp current load at 26 V DC. Two connector contacts shall be provided on the altimeter for this switching function signal.

3.4.11 Residual Height Adjustment - The equipment shall be adjustable to allow for installation of the Antennas in the aircraft from 2 to 20 feet above the ground. A weight-on-gear switch shall be used to actuate the "self-test" circuit whenever the airplane is on the ground; unless tests indicate that a reliable ground lock-on can be maintained during taxi.

3.4.12 Self-Test Provisions - A "GO, NO-GO" test provision shall be incorporated in the unit. This will be used for inflight and preflight testing of the equipment. Self-test shall be activated by depressing the control knob on the Indicator Unit or closing the contacts of the weight-on-gear switch and shall be capable of being energized continuously with no detrimental effects. The indicator shall indicate 5, +5, -10 feet whenever self-test is actuated.

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3.5 Detailed Requirements

3.5.1 Receiver-Transmitter RT-601B/APN-141(V) - The Receiver-Transmitter shall meet the following requirements.

3.5.1.1 Function - The Receiver-Transmitter shall transmit a radio frequency signal, receive the reflected signal, and derive the proper indication of radar altitude in accordance with the performance of this specification.

3.5.1.2 Form Factor - The Receiver-Transmitter shall be contained within the volume defined in Figure 1.

*3.5.1.3 Weight - The weight of the Receiver-Transmitter shall not exceed 6.5 pounds.

3.5.1.4 Contents of Unit - The Receiver-Transmitter shall contain the following assemblies and circuits.

3.5.1.4.1 Power Supply - The power supply shall convert the 115 V, 400 cps, single phase aircraft power to the AC and DC voltage outputs required for system operation.

3.5.1.4.2 Modulator - The modulator shall provide the triggering pulse to the transmitter cavity, the positive bias pulse and narrow pulse trigger to the microwave crystal switch, and a low-altitude fast ramp trigger pulse to the time comparator.

3.5.1.4.3 Transmitter Cavity - The transmitter cavity shall be a plate pulsed cavity oscillator having a manual frequency adjustment. The cavity is driven into oscillation by the modulator trigger pulse. A sample of the RF energy shall be coupled out of the cavity, detected and applied to the time comparator to operate the high altitude fast ramp. The output characteristics of the transmitter-cavity shall be as required for specified system operation.

3.5.1.4.4 Receiver Cavity - The receiver cavity shall be a superregenerative cavity fixed tuned to a specific frequency. Cavity oscillations are initiated when ground return signals are present coincident with the grid pulse from the time comparator.

3.5.1.4.5 Receiver - The receiver shall accept detected video pulses from the receiver cavity and produce tracking information and AGS control voltages. The receiver slow ramp output is supplied to the time comparator where it is compared to the fast ramp voltage to obtain proper receiver cavity pulse programming. The slow ramp output also controls the pulse width programmer in the RF Switching Unit. An altitude voltage is supplied to the Height Indicator where it is converted to an altitude pointer position.

3.5.1.4.6 Time Comparator - The time comparator shall provide ranging and timing information to other circuits in the system. Clock Pulse I, which is the primary timing pulse, operates the modulator transmit switch. Clock Pulse II controls the timing of the AGS pulse generator in the receiver. The time comparator supplies a pulse to the receiver cavity grid to actuate the receiver system.

*3.5.1.4.7 Reliability Circuit - The reliability circuit shall prevent the system from locking on noise, feedthrough, microphonic or other interfering or false signal. When the altitude signal strength is insufficient to produce reliable height indications, the reliability circuit shall cause the "OFF" flag to move into view on the Height Indicator within 0.5 second after loss of signal. When the signal strength becomes sufficient for reliable indication, the system shall lock on within 2.5 seconds. A reliability signal output having an impedance of 80,000 ohms ± 20% shall be provided at the electrical connector when the system is "locked on". The Height Indicator shall slew to the correct reading within 3.0 seconds after the system has "locked on" at true altitudes of 200 feet or more for the ID-881 and ID-1015 and at true altitudes of 300 or more for the ID-1304. A maximum of 5.0 seconds slewing time is allowable at true altitudes less than 200 feet for the ID-881 and ID-1015 and less than 300 feet for the ID-1304. When the altitude is in excess of 5,000 feet, the reliability circuit shall cause the altitude pointer to move behind the masked portion of the Height Indicator.

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3.5.1.5 Controls - There shall be no in-flight controls located in the Receiver-Transmitter.

*3.5.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type or Equivalent	Function
J101	77068-3068297 (PT02A-14-19-P)	Power Supply, Control Signals
J102	77068-3151266 (TNC)	RF Output to Switch Unit
J103	77068-3151265 (TNC)	Receiver Antenna
J104	77068-3068300 (TM)	Switch Gate

3.5.1.7 Mounting - The Receiver-Transmitter shall be hardmounted with mounting holes as shown in Figure 1.

3.5.1.8 Orientation - The Receiver-Transmitter shall be capable of operation to the accuracies required by this specification with no restriction in orientation.

3.5.2 RF Switching Unit, SA-791A/APN-141(V) - RF Switching Unit shall meet the following requirements.

3.5.2.1 Function - The RF Switching Unit shall provide a variable control of the transmitted pulse width. During the high altitude mode of operation, the RF Switching Unit shall connect the output of the transmitter cavity directly to the transmitting antenna.

3.5.2.1.1 The RF Switching Unit shall not require tuning.

3.5.2.2 Form Factor - The RF Switching Unit shall be contained within the volume defined in Figure 2.

*3.5.2.3 Weight - The weight of the RF Switching Unit shall not exceed 2.9 pounds.

3.5.2.4 Contents of Unit - The RF Switching Unit shall contain following assemblies and circuits.

3.5.2.4.1 RF Switching Circuits - The rf switching circuits shall provide the means to control the pulse width and amplitude of the transmitted rf energy during the low and high altitude modes of operation. The output and operation of the rf switching circuit shall have the characteristics required for specified system operation.

3.5.2.4.2 Drive Circuit - The drive circuit shall provide the narrow trigger pulse for the RF Switching Unit.

3.5.2.5 Controls - There shall be no in-flight controls located on the radio frequency Switching Unit.

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*3.5.2.6 Electrical Connections

Reference Designation	Receptacle Type or Equivalent	Function
J801	MS 3113H10-6P	Power Supply, Control Signals
J802	77068-3068515 (TNC)	RF Input from R/T
J804	77068-3068514 (TNC)	Switch Gate
J805	77068-3068515 (TNC)	Transmit Antenna

3.5.2.7 Mounting - The RF Switching Unit shall be hardmounted with mounting holes as shown in Figure 2.

3.5.2.8 Orientation - The RF Switching Unit shall be capable of operation to the accuracies required in this specification with no restriction in orientation.

*3.5.3 Indicator, Height ID-881B/APN-141(V), ID-1015A/APN-141(V) or ID-1304/APN-141(V) - The Indicator shall meet the following requirements.

3.5.3.1 Function - The Indicator shall indicate surface or terrain clearance by means of a single pointer against a fixed dial. A movable marker positioned against the same dial shall indicate the preselected height below which the low altitude limit switching function operates. Unreliable operation shall be indicated by an "OFF" flag coming into view when (1) the signal strength becomes inadequate to provide reliable altitude information, (2) the system is turned off or (3) aircraft power to the system is lost. Signal return at altitudes higher than 5,000 feet shall be indicated by movement of the pointer behind the masked portion of the Indicator dial.

*3.5.3.2 Form Factor - The Indicator shall be completely housed in a case similar to that illustrated in Figures 3, 4, or 5. Figure 3 is the outline drawing of a bezel mounted Indicator and Figure 4 is the outline drawing of a clamp mounted Indicator. Indicators per Figures 3 and 4 shall be interchangeable by changing the bezel ring. Figure 5 is the outline drawing of the ID-1304/APN-141(V).

*3.5.3.3 Weight - The weight of the ID-881B/APN-141(V) or ID-1015A/APN-141(V) shall not exceed 2.1 pounds. The weight of the ID-1304/APN-141(V) shall not exceed 2.3 pounds.

3.5.3.4 Contents of Unit - This unit shall include the necessary mechanism to rotate the altitude indicator pointer to a position determined by the system. It shall also include a low altitude limit index.

*3.5.3.5 Controls - A control knob located as shown in Figures 3, 4, and 5 shall control the low altitude limit index, actuation of the self-test circuit, and also function as the power control switch. The low altitude limit index shall be continuously adjustable throughout the range of the equipment. When the equipment is turned off this marker shall be located under the masked portion of the indicator. When the equipment is turned on the initial position of the limit index shall be at or below 0 feet.

*3.5.3.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type or Equivalent	Function
J601	77820-PTIH-14-18P	Power Supply, Control Signals

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A single multicontact connector shall be mounted on the back of the Indicator opposite the dial for all electrical connections as shown in Figures 3, 4, and 5.

*3.5.3.7 Dial Markings - The dial of the ID-881B/APN-141(V) and ID-1015A/APN-141(V) shall be similar to that shown in Figure 6. The dial of the ID-1304/APN-141(V) shall be similar to that shown in Figure 7. The tip of the pointer shall coincide with the innermost point of the minor graduations. The low altitude limit index shall be in the form of an equilateral triangle and shall be positioned to provide 0.020 inch clearance with the pointer tip.

3.5.3.8 Indicator Protection - Protection shall be provided to avoid damaging the Indicator at altitudes above 5,000 feet and the system shall resume normal operation while descending through 5,000 feet.

3.5.3.9 Instrument Lighting - Indicators shall be integrally lighted in compliance with Specification MIL-L-25467. The maximum limit of brightness requirement does not apply to the ID-1304/APN-141(V) limit index.

*3.5.3.10 External "OFF" Flag Control - The ID-881B Indicator shall contain provisions for external control of the "OFF" flag.

3.5.4 Antenna AT-1015/APN-141(V) or AS-1233/APN-141(V) - Two identical antennas shall be provided for this system. The antennas shall meet the requirements of the system and this specification as follows.

3.5.4.1 Form Factor - The Antennas shall be contained within the volumes defined in Figures 8 and 9.

*3.5.4.2 Weight - The weight of each Antenna shall not exceed 0.25 pound.

3.5.4.3 Electrical Connections

Reference Designation	Receptacle Type or Equivalent	Function
J701	77068-1079568 (TNC)	RF Signal from Transmitter or RF Signal to Receiver

3.5.4.4 Mounting - The Antenna shall have mounting provisions as shown in Figures 8 or 9. The Antenna shall withstand a differential pressure of one atmosphere.

3.5.4.5 Orientation - System accuracy requirements shall be met with Antenna spacing of 18 inches to 60 inches center to center.

3.5.4.6 Antenna Pattern - Performance requirements shall be met with Antenna pattern interferences of particular aircraft design for which the altimeter set is intended.

*3.5.5 Coupler, Linearizer CU-1464/APN-141(V) - The Coupler shall meet the following requirements.

*3.5.5.1 Function - When operated in conjunction with RT-601B/APN-141(V), the Coupler shall provide to the using equipment a linearized altitude signal output having a 0.1 second time constant and a one second memory.

*3.5.5.2 Form Factor - The Coupler shall be contained within the volume defined in Figure 10.

*3.5.5.3 Weight - The weight of the Coupler shall not exceed 1.5 pounds.

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*3.5.5.4 Operation of Unit

*3.5.5.4.1 Altitude Signal - The Coupler shall be used in conjunction with the RT-601B/APN-141(V) to provide a linearized altitude signal from 0 to 5000 feet with an output impedance of one ohm maximum, 5000 ohm load minimum. The linearized altitude signal shall follow the input altitude signal with a 0.1 second time constant. Upon loss of "track" signal from the RT Unit, the linearized output from the Coupler shall maintain the last input altitude for a period of one second. If at the end of one second, the RT Unit has not regained "track," the output signal voltage from the Coupler shall go to zero. However, if the RT Unit regains "track" during the one second memory period, there shall be no indication provided by the Coupler to the using equipment that a momentary loss of "track" had occurred.

*3.5.5.4.2 Reliability Signal - The reliability signal from the RT Unit shall be processed and delayed by one second in the Coupler with the following output characteristics: "Search" condition = 0 volts; "Track" condition = 5 volts; output impedance = 1.5 K ohms maximum.

*3.5.5.4.3 Indicator Failure Protection - The Coupler shall contain provisions to prevent normal failures of the Altimeter Indicator Unit from causing the coupler output to be incorrect without also causing the coupler reliability signal to switch to the unreliable "Search" condition.

*3.5.5.5 Altitude Accuracy - The Coupler shall provide an output of +5 mv/foot from 0 to 5000 feet when operated in conjunction with the RT-601B/APN-141(V) with an accuracy of ±7 feet or ±7% whichever is greater.

*3.5.5.6 Controls - There shall be no in-flight controls located on the Coupler.

*3.5.5.7 Electrical Connections - Connection to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J901	MS 3113H12-10P	Power Supply, Control Signals

*3.5.5.8 Mounting - The Coupler shall be hardmounted with mounting holes as shown in Figure 10.

*3.5.5.9 Orientation - The Coupler shall be capable of operation to the accuracies required by this specification with no restriction in orientation.

*3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see paragraph 6.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

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(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

*4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed in the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the government; it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

*4.3 Production Sample Tests - One of the first 10 production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for production sample tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period.

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The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Production Sample Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production sample, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by Specification MIL-R-23094. Equipments selected for reliability assurance tests shall first have passed the individual tests.

*4.4.3.1 Reliability Procedure - For the reliability Qualification Plan, as outlined in Specification MIL-R-23094, Procedure I shall be used. For the Reliability Sampling Plan Procedure II shall be used.

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4.4.3.2 Test Level - The Test Level shall be III from Table I of Specification MIL-R-23094.

*4.4.3.3 Accept-Reject Criteria - Figure 13 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the Reliability Qualification Test Plan. Figure 15 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the Reliability Sampling Test Plan.

4.4.3.4 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 3 hours during each cycle.

4.4.3.5 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified in 6.9.

4.4.3.6 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the Acceptance requirement (paragraph 4.4.3.5) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.7 Preventive Maintenance - During the period of the tests the following preventive maintenance measures may be performed upon the equipment:

Measure	Time
Readjust Equipment	Every 50 hours

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

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(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

*4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following: (Equipments which have successfully passed the production sample test, sampling tests, reliability tests, or special tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ±5 volts (at lowest applicable frequency)
A. C. or D. C. Voltage	5.0 ±0.5 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

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(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The Electronic Altimeter Set AN/APN-141(V), covered by this specification is intended to be installed in aircraft of varying designs and configurations. Installation in a particular type of aircraft is dependent upon satisfactory flight evaluation tests using the type of aircraft for which the production altimeter is intended.

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6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 5.0 ±0.5 V (AC or DC) and 115 ±1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Non-repairable Subassemblies - As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

*6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 3.6).

(4) Type of height Indicator and Antenna required or Linearizer Coupler, if required (see 6.8).

6.7 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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*6.8 Items of Equipment - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Applicable Paragraph
Receiver-Transmitter, Radar	RT-601B/APN-141(V)	3.5.1
Switching Unit, Radio Frequency	SA-791A/APN-141(V)	3.5.2
Indicator, Height	ID-881B/APN-141(V) or ID-1015A/APN-141(V) or ID-1304/APN-141(V)	3.5.3
Antenna	AT-1015/APN-141(V) or AS-1233/APN-141(V)	3.5.4
Coupler, Linearizer	CU-1464/APN-141(V)	3.5.5.

6.9 Performance Characteristics to be Measured

*6.9.1 Altitude Accuracy - ±5 feet or ±5% whichever is greater, (±7 feet or ±7% for coupler), measured at approximately 10, 100, 300, 1000 and 4000 ft. Accuracy to be determined by the altitude voltage, as illustrated in the following chart:

Simulated Altitude	Indicator Reading Limits	Altitude Voltage Limits	Coupler Voltage Limits
10 ft	5 - 15 ft	1.10 - 1.30V	0.015 - 0.085V
100 ft	95 - 105 ft	2.90 - 3.10V	0.465 - 0.535V
300 ft	285 - 315 ft	6.70 - 7.30V	1.395 - 1.605V
1000 ft	950 - 1050 ft	10.43 - 10.70V	4.650 - 5.350V
4000 ft	3800 - 4200 ft	17.87 - 18.91V	18.600 - 21.400V

6.9.2 Reliability Signal

Unit	"Search"	"Track"
RT	0 to -1.5 Volts	4 ±0.5 Volts
Coupler	0 ±0.5 Volt	5 ±0.5 Volts

6.9.3 Sensitivity - System must remain in "track" mode (locked on to simulated target) with the following loop attenuations:

Simulated Altitude	Total Attenuation
100 ft	90 db
4000 ft	121 db

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6.10 Applicable Documents -

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment; Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MS17322	Meter, Time Totalizing, Miniature, Digital, 115 Volt, 400-Cycle

PUBLICATIONS

Bureau of Naval Weapons

EI-425	Avionics Installation Instructions for Altimeter Set, Electronic AN/APN-141(V)
ET-425	Avionics Bench, Preflight and Flight Test Instructions for Installation of Altimeter Set, Electronic AN/APN-141(V)

6.11.1 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

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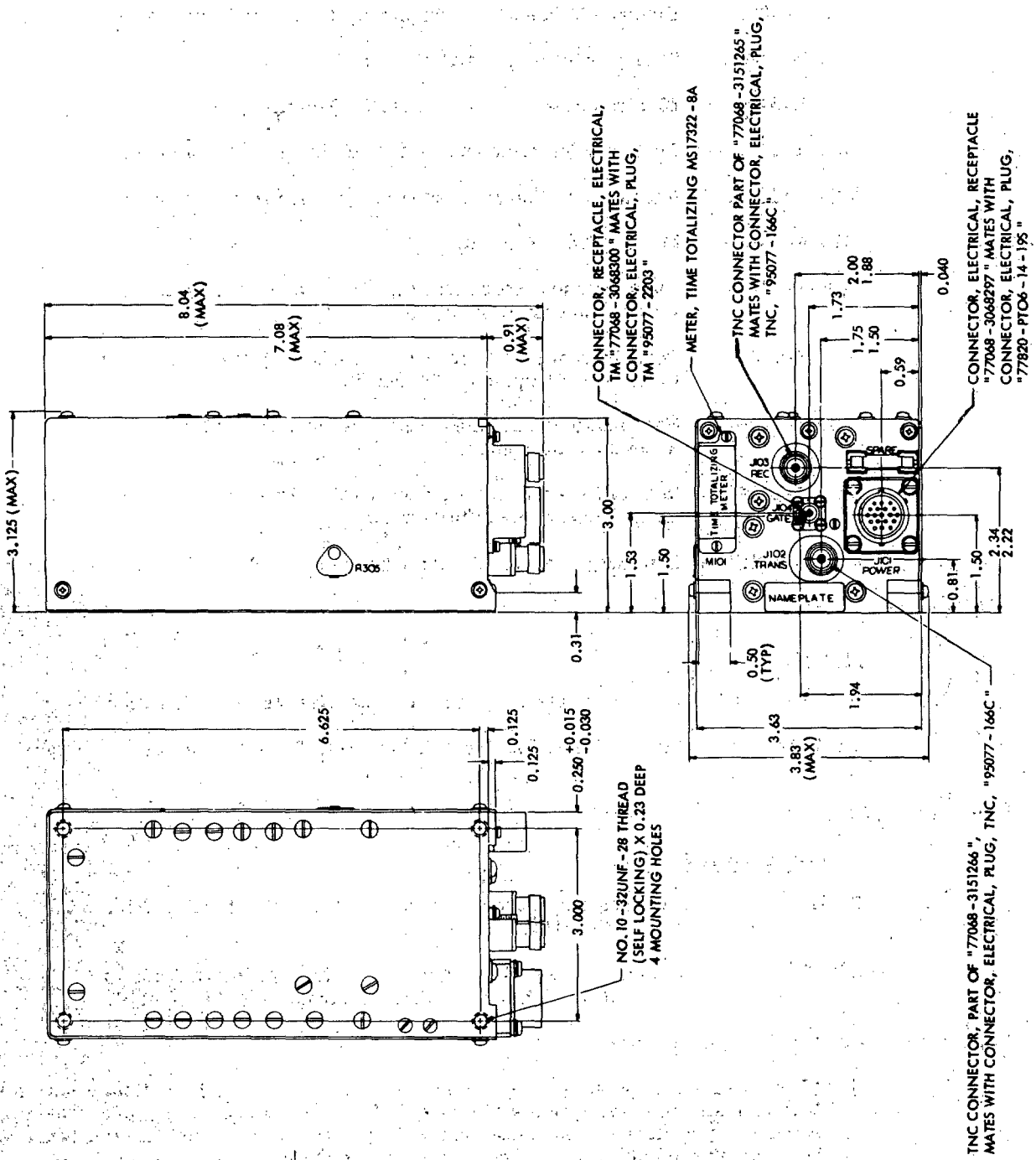


Figure 1. Receiver-Transmitter RT-601B/APN-141(V)

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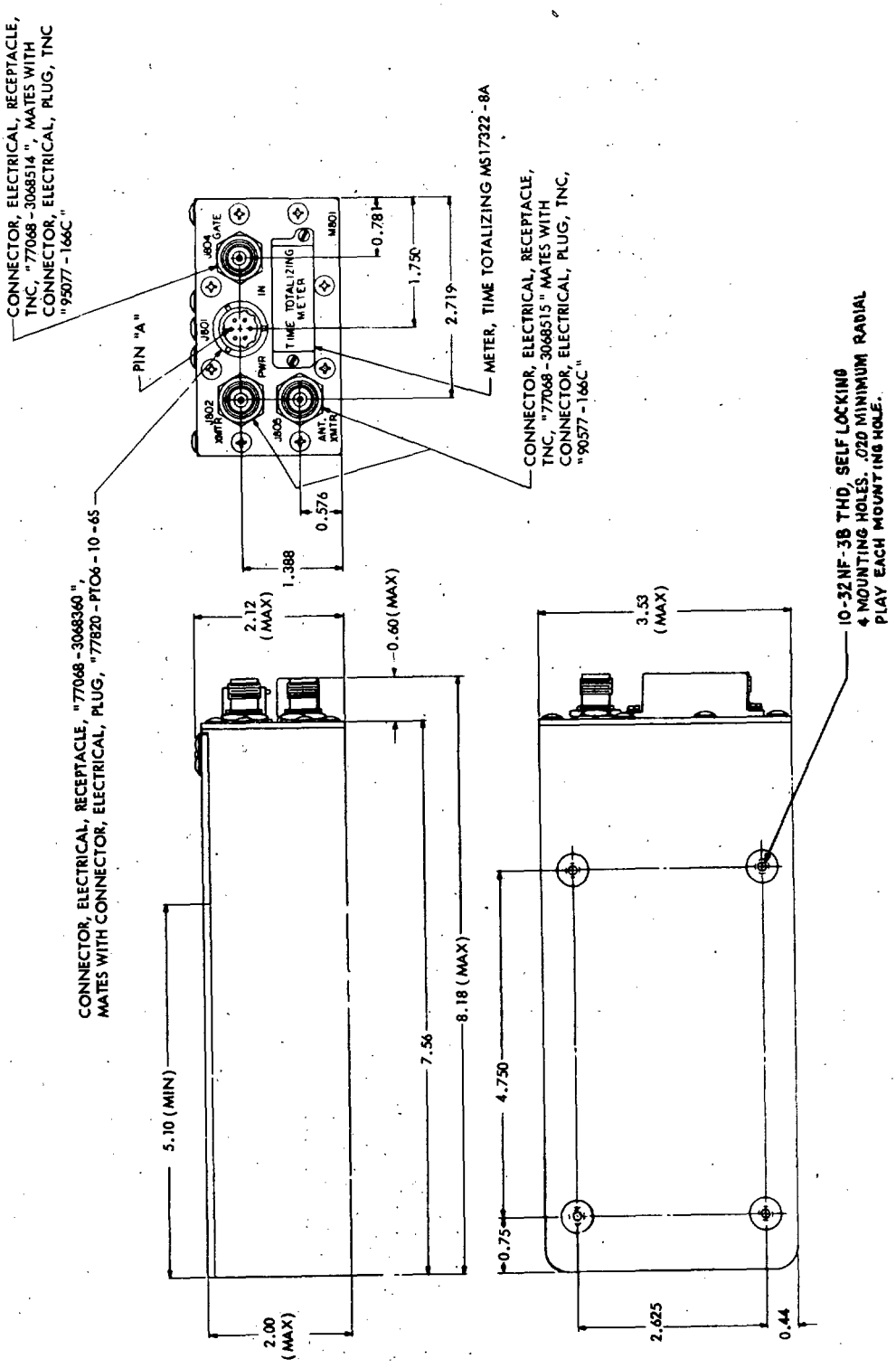


Figure 2. RF Switching Unit SA-791A/APN-141(V)

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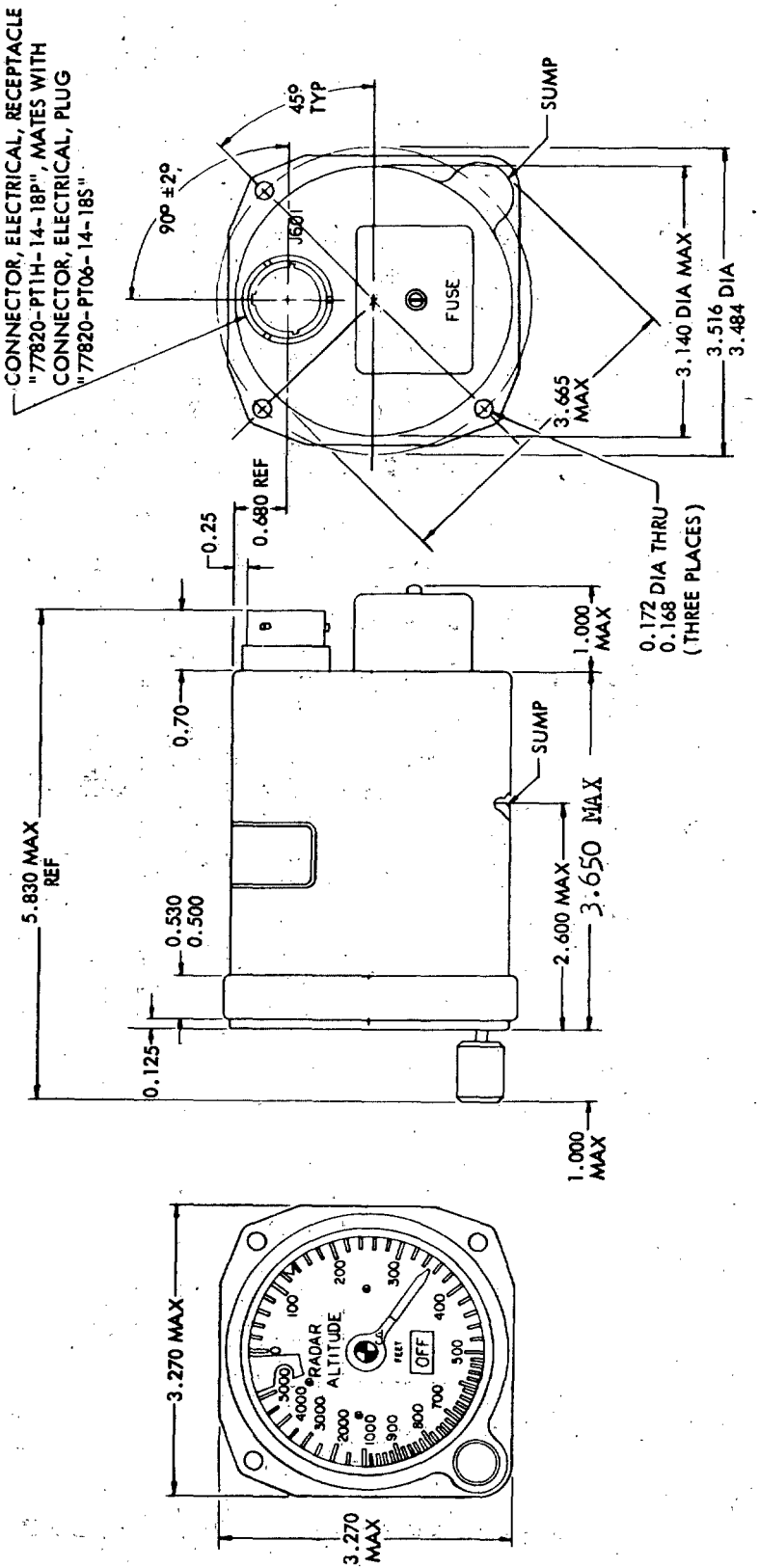


Figure 3. Indicator ID-881B/APN 141(V)

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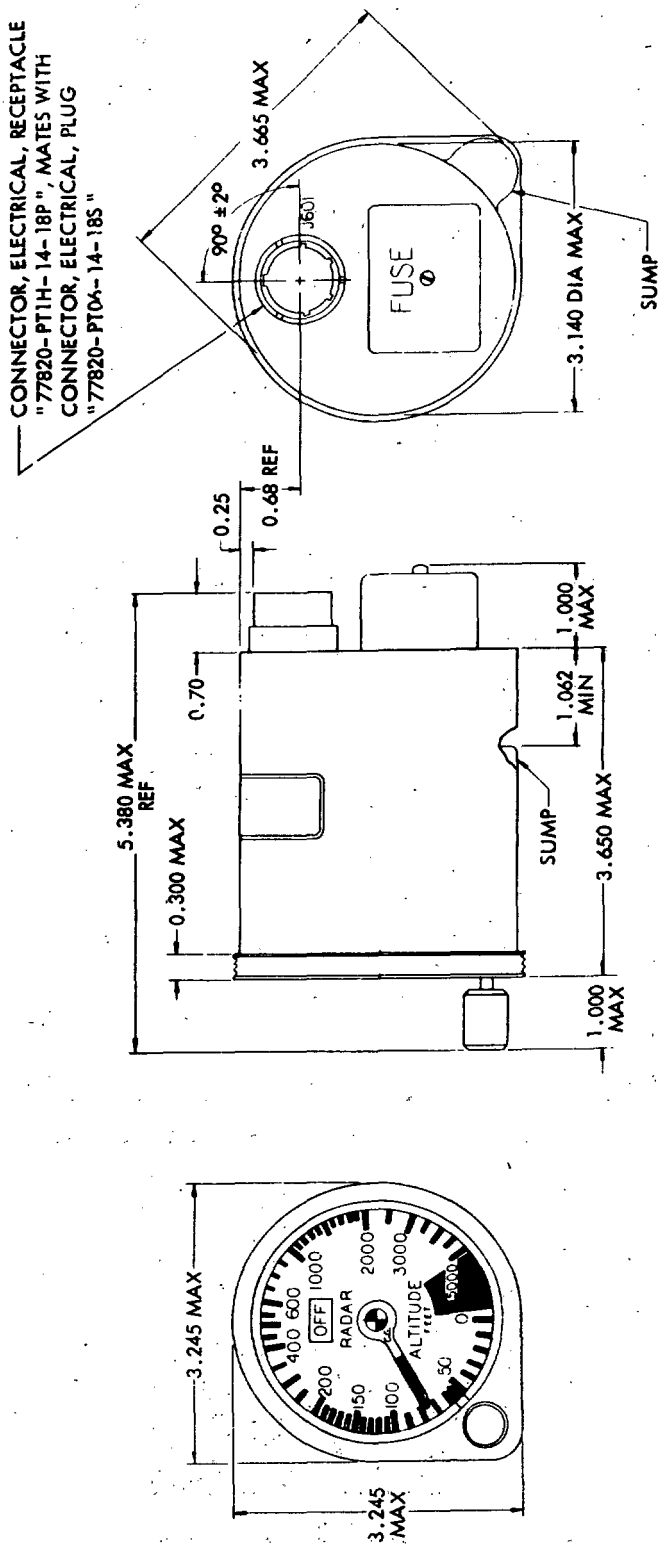


Figure 4. Indicator ID-1015A/APN-141(V)

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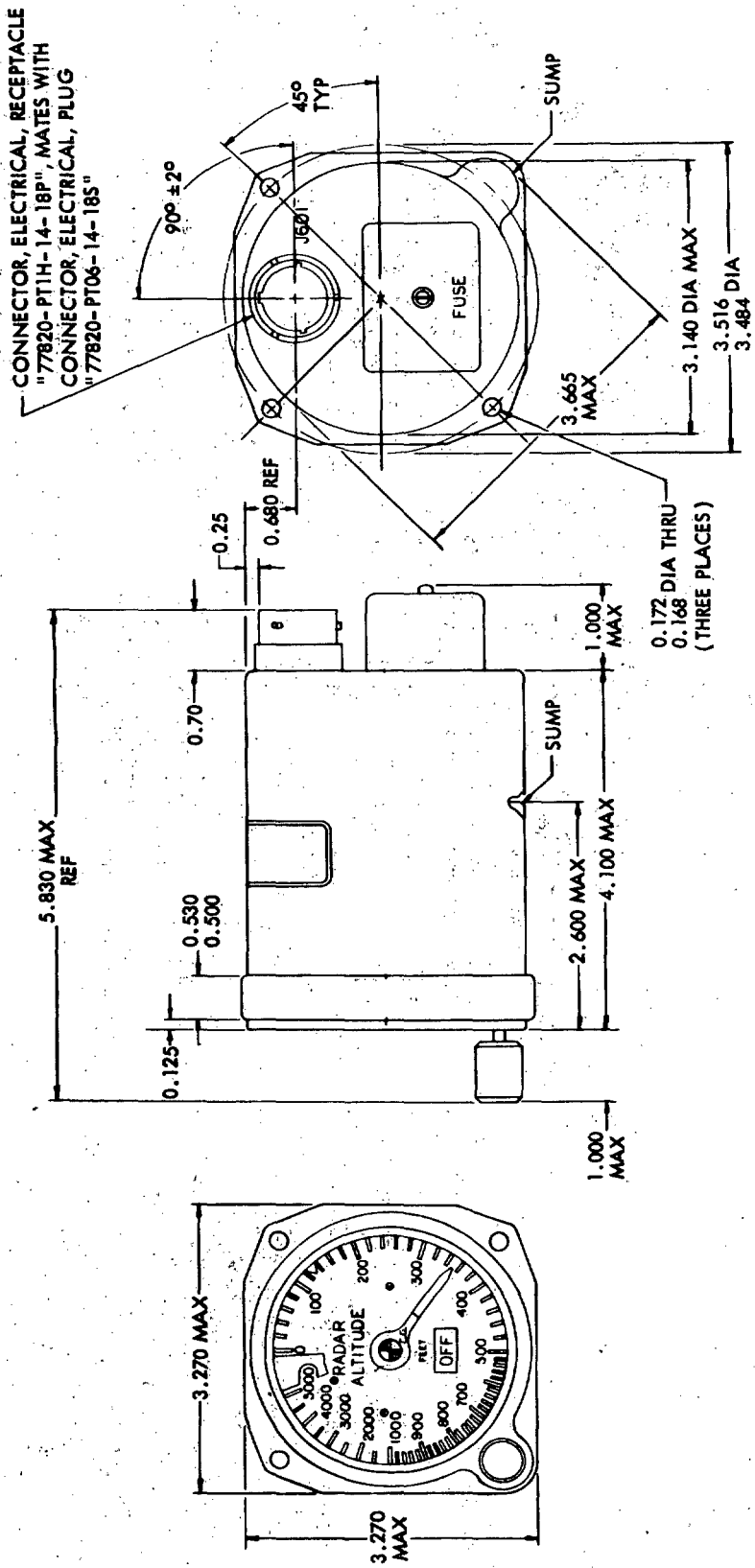
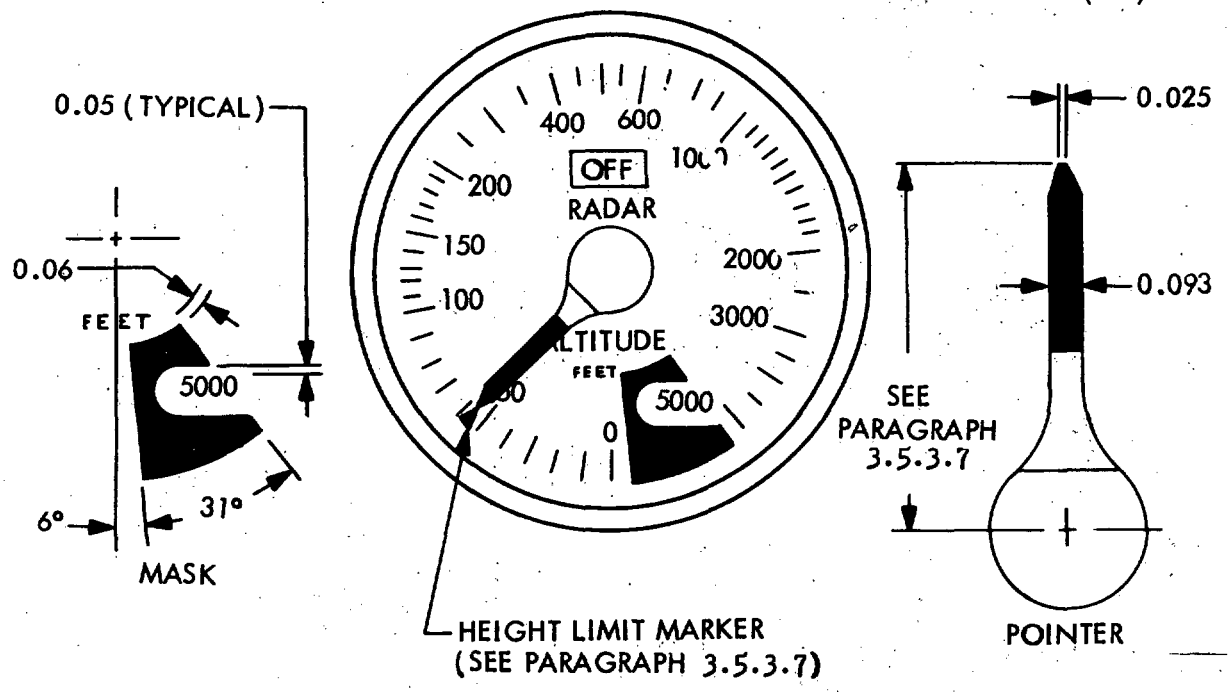


Figure 5. ID-1304/APN-141(V) Indicator

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GRADUATIONS		Length ±.01	Width ±.005
Increment	Range		
100'	600' - 2000'	.100	.025
10'	0' - 200'		
50'	200' - 600'	.160	.030
500'	1000' - 5000'		
50'	0' - 200'		
100'	100' - 600'	.220	.035
1000'	1000' - 5000'		

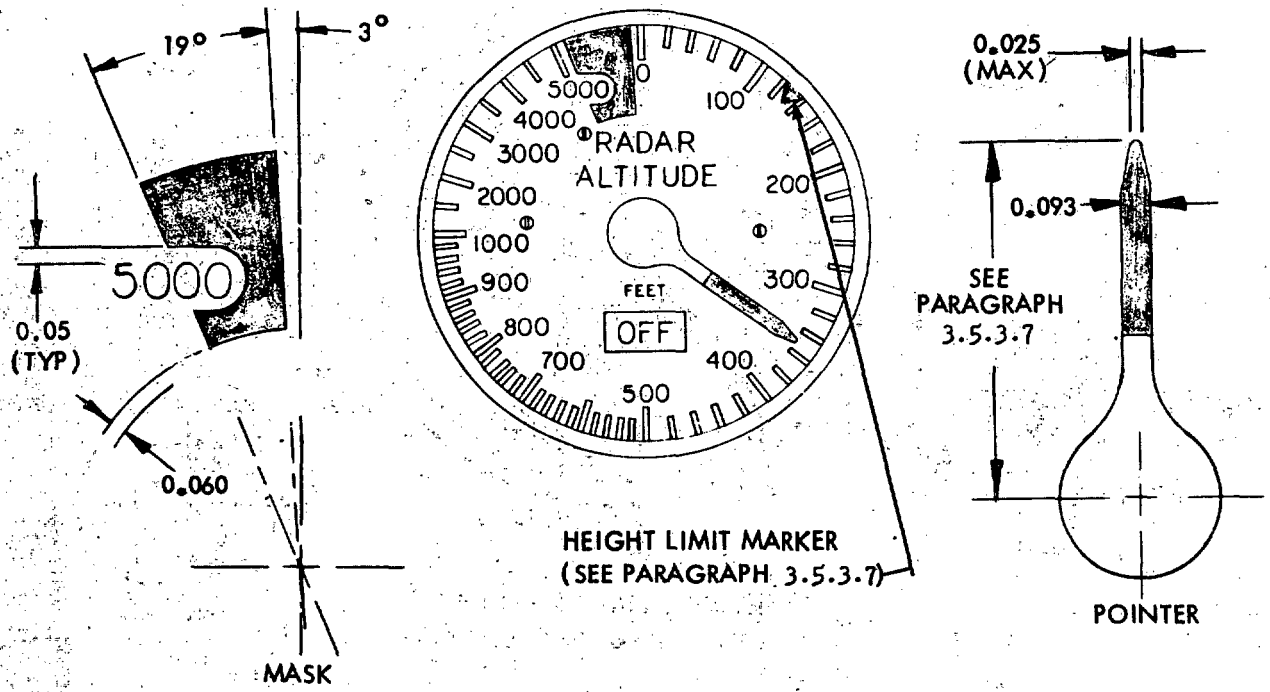
Calibration Table (Major Graduations)
Degrees Clockwise ±15' from 6 o'clock

Feet	Degrees	Feet	Degrees
0	0°	1000	221°15'
50	40°	2000	263°45'
100	80°	3000	288°42'
150	101°15'	4000	304°21'
200	122°30'	5000	320°
300	143°45'		
400	165°		
500	177°30'		
600	190°		

MARKINGS:	HEIGHT ±.01
Numerals except "5000"	.150
"Radar", "Altitude"	.150
"Feet"	.060
"5000"	.100
"OFF" (Black on White)	.150

Figure 6. ID-881B and ID-1015A - Indicator Dial Configuration

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GRADUATIONS:

Increment	Range	Length ±0.01	Width ±0.005
20'	0' - 1000'	0.160	0.030
500'	1000' - 5000'		
100'	0' - 1000'	0.220	0.035
1000'	1000' - 5000'		

Calibration Table (Major Graduations)
Degrees Clockwise ±15' from 0 Ft. (0 Ft. = 12 O'Clock ±1°)

Feet	Degrees	Feet	Degrees
0	0	800	233.10
100	35.35	900	250.80
200	70.75	1000	268.50
300	106.15	1500	276.80
400	141.50	2000	285.10
500	180.00	3000	301.75
600	197.70	4000	318.40
700	215.40	5000	335.00

Markings	Height ±0.01
Numerals	0.100
"RADAR" "ALTITUDE"	0.120
"FEET"	0.060
"OFF" (Black on White)	0.150

Figure 7. ID-1304/APN-141(V) Indicator Dial Configuration

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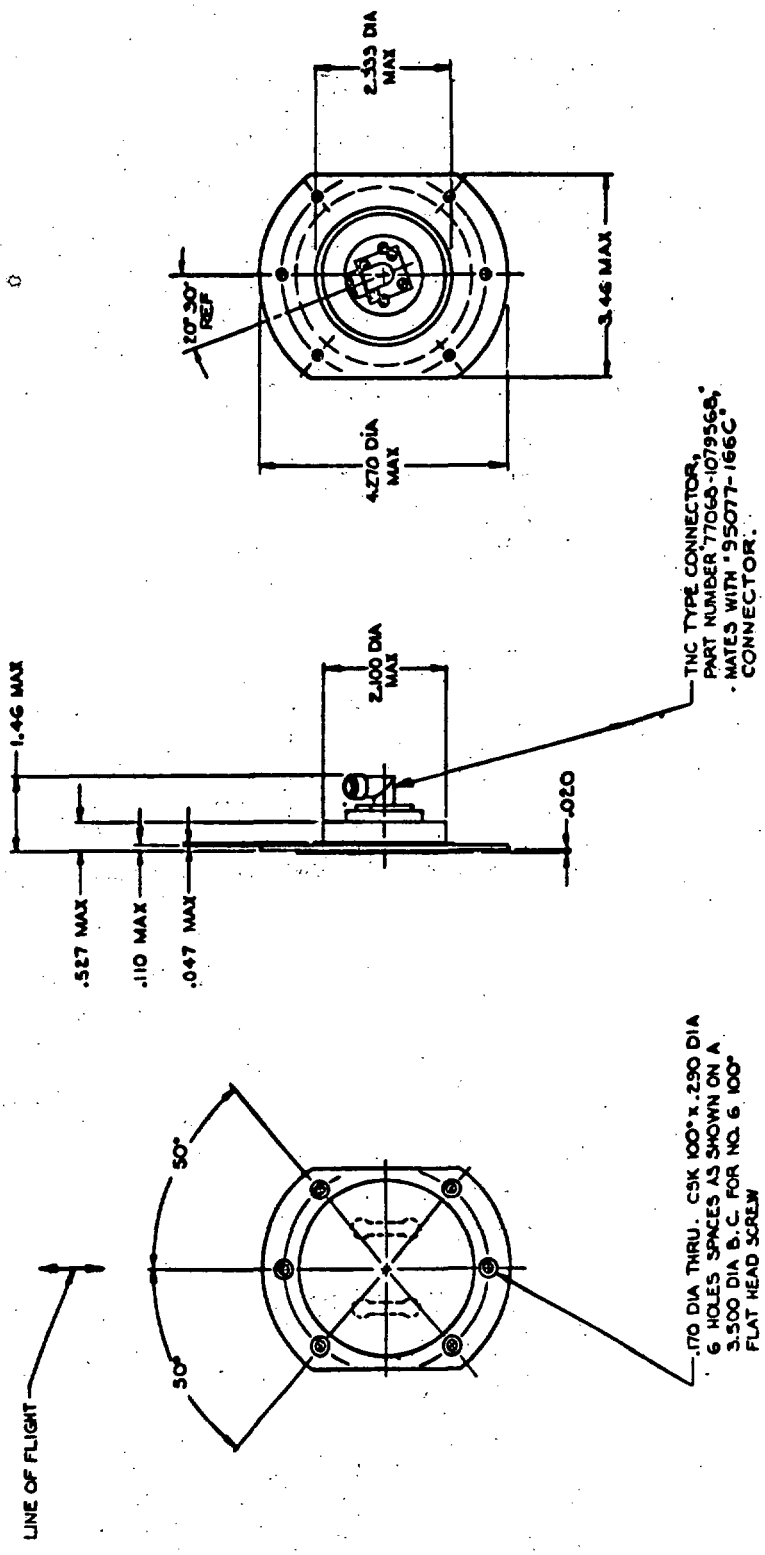


Figure 8. Antenna AS-1233/APN 141(V)

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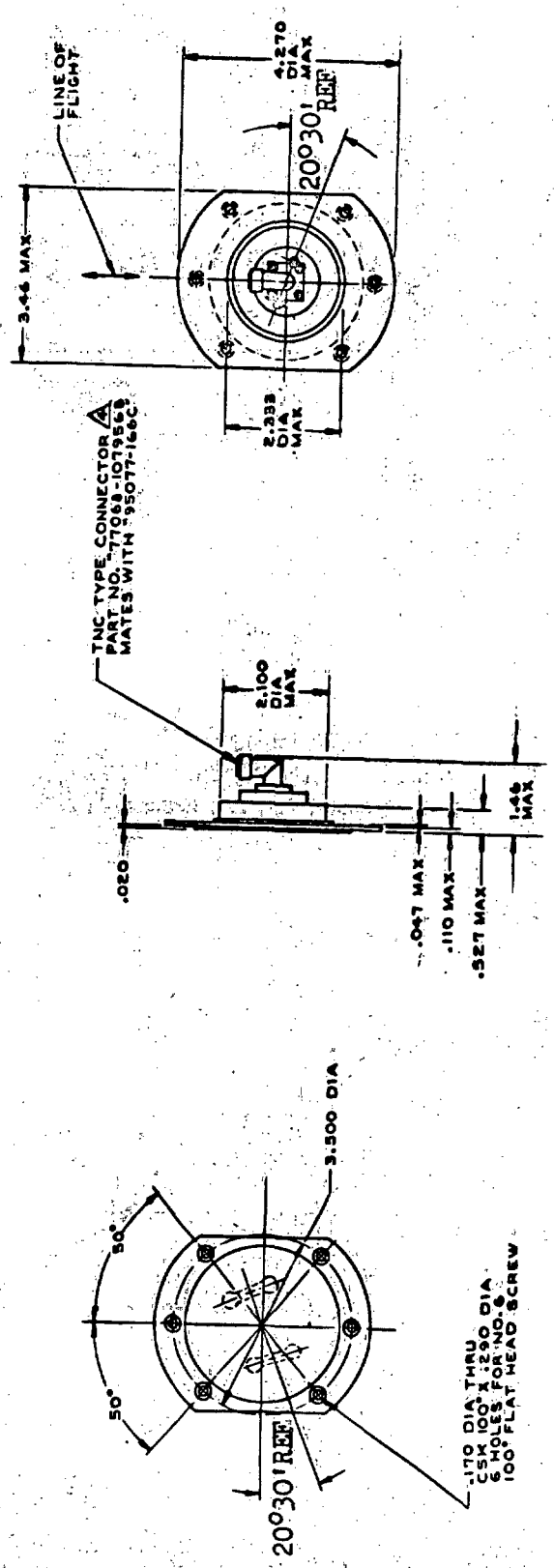


Figure 9. Antenna AT-1015/APN-141(V)

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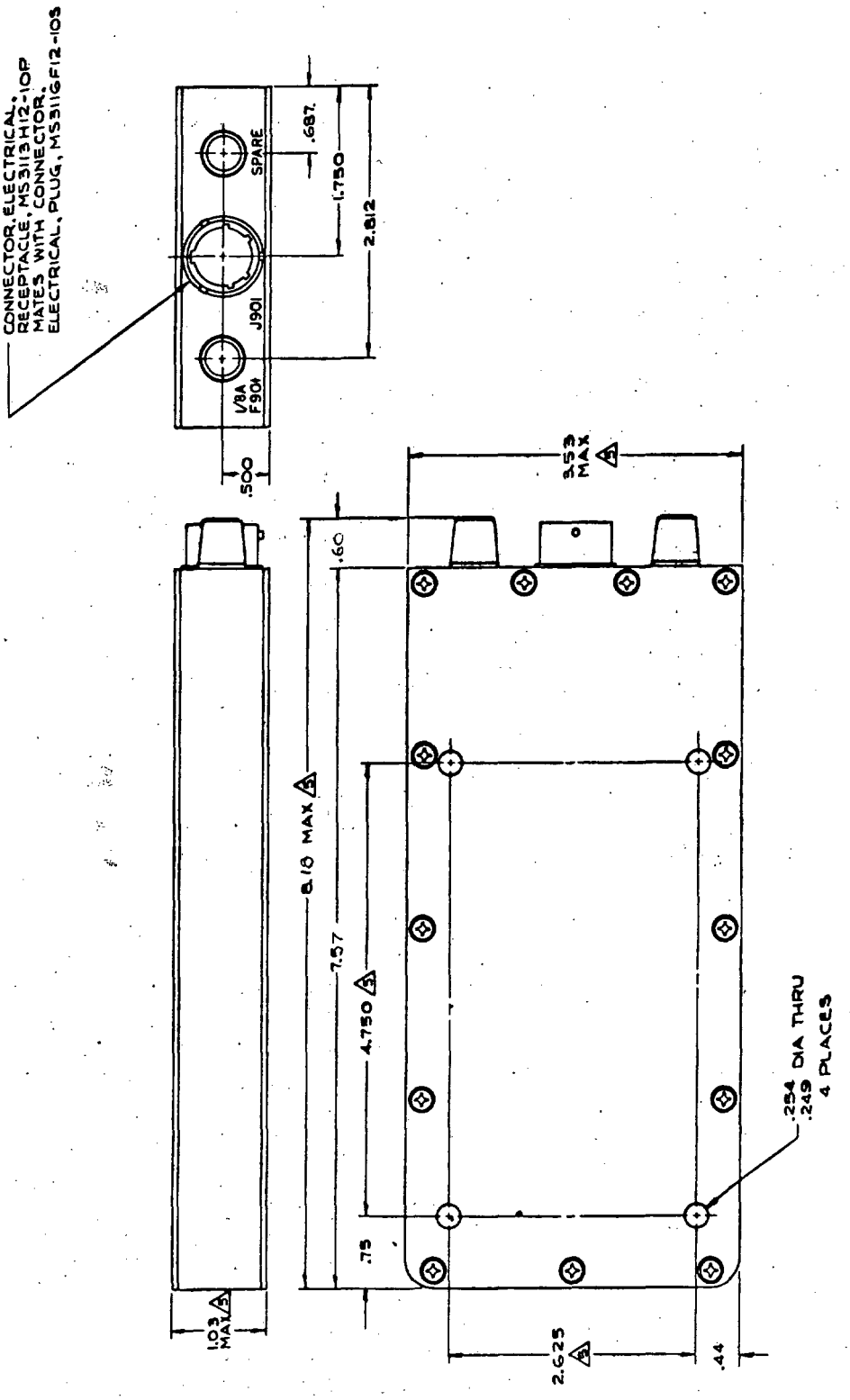


Figure 10. Coupler, Linerizer CU-1464/APN-141(V)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-A-23887A(WP) ALTIMETER SET, ELECTRONIC AN/APN-141(V)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8279

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